

Program: Diploma in Microelectronics	
Course Code: 4492	Course Title: Introduction to MEMS
Semester: 4	Credits: 3
Course Category: Program Core	
Periods per week: 3 (L:3, T:0, P:0)	Periods per semester: 45

Course Objectives:

- To understand the operation of major classes of MEMS devices/systems
- To know about the various materials used for MEMS
- To grasp the fundamentals of standard micro fabrication processes
- To understand the basics of micromanufacturing and micro system packaging

Course Prerequisites:

Topic	Course code	Course Title	Semester
Polymers, Oxidation, Electroplating		Applied Chemistry	1
Passive components		Fundamentals of Electrical & Electronics Engineering	2
Fabrication processes		Introduction to IC Fabrication	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Describe the working principles of micro sensors and actuators	12	Understanding
CO2	Identify the typical materials used for fabrication of micro systems	10	Understanding
CO3	Explain the processes involved in micro fabrication	11	Understanding
CO4	Understand the principles of standard micro fabrication techniques and packaging	10	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	2				
CO2	3		2				
CO3	3		2				
CO4	3		2				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the working principles of micro sensors and actuators		
M1.01	Explain the components of a microsystem	2	Understanding
M1.02	Discuss the working of thermal sensors and actuators	4	Understanding
M1.03	Explain the operation of electrostatic sensors and actuators	4	Understanding
M1.04	Describe the working of piezoelectric sensors and actuators	2	Understanding
Contents: Overview of MEMS and Microsystems MEMS and Microsystems - MEMS as a microsensor, MEMS as a microactuator, Components of a microsystem, Intelligent microsystem, Multidisciplinary nature of MEMS, Applications Sensors and Actuators : Thermal sensors - Thermocouple, Thermistor, Thermal actuators - Thermal Bimorph, Microvalve, Actuation using SMA Electrostatic sensors - Microaccelerometer, Electrostatic actuators - Microgripper, Rotary Micromotor, Micropump Piezoelectric sensors - Piezoelectric accelerometer, Piezoelectric actuators - Actuation using Piezoelectric crystal			
CO2	Identify the typical materials used for fabrication of micro systems		

M2.01	Outline the properties of silicon for MEMS application	2	Understanding
M2.02	Explain the different silicon compounds used in fabrication process	4	Understanding
M2.03	Discuss about GaAs and piezoelectric crystals as MEMS materials	2	Understanding
M2.04	Explain the role of polymers and LB film in MEMS	2	Understanding
	Series Test - I	1	

Contents:

Materials for MEMS and Microsystems

Substrates and Wafers, Silicon as an ideal substrate for MEMS, Mechanical Properties of silicon, Silicon compounds – Silicon Nitride, Silicon Dioxide, Silicon carbide, Poly Silicon Silicon Piezo resistors, GaAs, Quartz, Piezoelectric Crystals, Polymers, Langmuir – Blodgett Films.

CO3	Explain the processes involved in micro fabrication		
M3.01	Illustrate the photolithographic process and ion implanatation process	3	Understanding
M3.02	Describe the process of diffusion and oxidation in MEMS fabrication	3	Understanding
M3.03	Explain the vapour deposition techniques involved in micro system fabrication	3	Understanding
M3.04	Discuss epitaxy and etching techniques	2	Understanding

Contents:

Micro system Fabrication Processes

Photolithography – Ion implantation – Diffusion – Oxidation – Chemical Vapour Deposition – Physical Vapour Depostion – Deposition of Epitaxy - Etching

CO4	Understand the principles of standard micro fabrication techniques		
M4.01	Discuss Bulk Micromanufacturing and Surface Micromachining	2	Understanding
M4.02	Explain LIGA Process and Microstereo lithography	2	Understanding
M4.03	Discuss the considerations in packaging design and levels of Micro system packaging	3	Understanding
M4.04	Describe the bonding techniques for MEMS	3	Understanding

	Series Test - I	1	
Contents: Overview of Micromanufacturing and Micro system Packaging Bulk Micromanufacturing, Surface Micromachining, The LIGA Process, Microstereo lithography Micro system Packaging - General considerations in packaging design, Levels of Micro system packaging Bonding techniques for MEMS: Surface bonding, Anodic bonding, Silicon - on - Insulator, wire bonding			

Text /Reference:

T/R	Book Title/Author
T1	Tai-Ran Hsu, MEMS and Microsystems Design and Manufacture, TMH, 2002
T2	Chang Liu, Foundations of MEMS, Pearson 2012
R1	Stephen D. Senturia, Microsystem design, Springer (India), 2006
R2	Julian W Gardner, Microsensors: Principles and Applications, John Wiley & Sons, 1994
R3	Mark Madou, Fundamentals of Micro fabrication, CRC Press, New York, 1997

Online Resources:

Sl. No	Website Link
1	https://youtube.com/playlist?list=PLgMDNELGJ1CbufZjqWa8uoSlQWKqVwPN7&si=GGoirly9vS3pqc_p
2	https://youtube.com/playlist?list=PL5873EDBDFB69BAD8&si=qg8uWLk5pfxCXzyL
3	https://terpconnect.umd.edu/~sandborn/research/JPL_MEMS/microeng_liga.html
4	https://nptel.ac.in/courses/117105082